

TD N° 2. Constraint satisfaction problems (CSP)

Problem solving-2-

Exercise 1: Problem formulation

- a) Consider a **Sudoku** puzzle as a CSP. Define the variables, domains, and constraints for this problem.

5	3			7			
6			1	9	5		
	9	8				6	
8				6			3
4			8		3		1
7				2			6
	6					2	8
			4	1	9		5
				8		7	9

Define the variables, domains, and constraints for this problem.

b) -CLASS SCHEDULING

- Student Ahmed needs 4 more required classes to graduate

A: Advanced Algorithms **B:** Bayesian Learning

C: Computer Programming **D:** Distributed databases

- **Some restrictions**

Advanced Algorithms must be taken same semester as Distributed databases

Computer programming is a prerequisite for Distributed databases and Bayesian learning, so it must be taken in an earlier semester

Advanced algorithms and Bayesian Learning are always offered at the same time, so they cannot be taken the same semester

- 3 semesters (semester 1, 2, 3) when can take classes

- c) Consider a cryptarithmic problem where you need to assign digits to letters to solve an arithmetic equation.

For example,

-Define the variables, domains, and constraints for this problem.

$$\begin{array}{r} \text{S E N D} \\ + \text{M O R E} \\ \hline \text{M O N E Y} \end{array}$$

Exercise 2:

- 1) -Use Backtracking and solve Class scheduling
- 2) Consider the following CSP model:

- Variables : X, Y, Z
- Domain : $d = \{1, 2, 3, 4\}$ for each variable
- Constraints :
 1. $X + Y = Z$
 2. $X > Y$
 3. $Z < 4$

-Use backtracking algorithm to find a solution for this CSP. Show the search tree and the solution if exists.

Exercise 3:

You are responsible for scheduling professors for computer science classes that take place on **Mondays, Wednesdays, and Fridays**. There are **5** classes that must be taught and **3** professors available. Each professor can teach only one class at a time, so a professor cannot be assigned to two classes whose times overlap.

The classes and their schedules are the following:

- Class 1 – Introduction to Artificial Intelligence: 8:00 – 9:00
- Class 2 – Networks: 8:30 – 9:30
- Class 3 – Theory of Languages: 9:00 – 10:00
- Class 4 – Operations Research: 9:00 – 10:00
- Class 5 – Security: 9:30 – 10:30

The professors are:

- Professor A, who is available to teach Classes 3 and 4.
- Professor B, who is available to teach Classes 2, 3, 4, and 5.
- Professor C, who is available to teach Classes 1, 2, 3, 4, and 5.

Task:

Formulate this problem as a **Constraint Satisfaction Problem (CSP)** by identifying the **variables, domains, and constraints**. Draw the constraint graph associated with your CSP apply arc consistency and provide one solution .

Exercise 4

Find a four-digit number satisfying the following conditions:

1. the number is odd,
2. the number only contains the digits 1, 2, 3, 4, and 5
3. each digit (except the leftmost) is strictly larger than the digit to its left.

Exercise 5

Consider the following constraint satisfaction problem (CSP):

Variables: X_1, X_2, X_3

Domains: $D(x_1) = \{1, 2, 3, 4\}$, $D(x_2) = \{a, b, c\}$, $D(x_3) = \{\alpha, \beta, \gamma\}$,

Constraints:

$$C(X_1, X_2) = \{\langle 1, a \rangle, \langle 2, b \rangle, \langle 3, a \rangle, \langle 3, b \rangle, \langle 4, b \rangle\}$$

$$C(X_1, X_3) = \{\langle 1, \beta \rangle, \langle 3, \beta \rangle, \langle 4, \beta \rangle\}$$

$$C(X_2, X_3) = \{\langle a, \gamma \rangle, \langle b, \beta \rangle, \langle b, \alpha \rangle, \langle c, \gamma \rangle\}$$

Solve the problem using:

1. backtracking;
2. backtracking and forward checking;
3. backtracking with minimum-remaining-values heuristic and forward checking;
4. backtracking with minimum-remaining-values and least-constraining-value heuristics and forward checking